

Third Terminal Examination – 2082

Class : XI

Subject: Physics (1011)

Full Marks: 75**Time: 3 hrs.**

SET 'A'

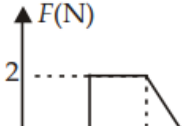
Pass Marks: 30

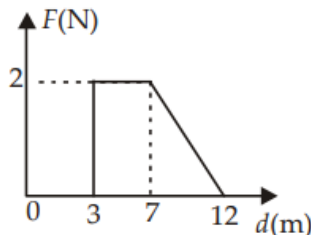
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- Dimensional formula of Specific Heat Capacity is:
a. $[M^0 L^1 T^{-2}]$
b. $[M^0 L^2 T^{-2}]$
c. $[M^0 L^2 T^{-2} K^{-1}]$
d. $[M^0 L^2 T^{-2} K^{-2}]$
 - The sum of two unit vectors is a unit vector. Then their difference will be:
a. $\sqrt{2}$
b. 2
c. $\sqrt{3}$
d. Zero
 - Silviya goes to her school at a speed of 30 km/hr and returns back at 60 km/hr along the same path. Her average speed is:
a. 35 km/hr
b. 40 km/hr
c. 45 km/hr
d. 50 km/hr
 - A machine gun fires a bullet of mass 40 g with a velocity 1200 m/s. The man holding it can exert a maximum force of 144 N on gun. How many bullets can he fire per second at the most?
a. 1
b. 4
c. 2
d. 3
 - Force 'F' on a particle moving in a straight line varies with distance 'd' as shown in figure. The work done on the particle during its displacement from 3 m to 12 m is:
a. 18 J
b. 21 J
c. 26 J
d. 13 J
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- The graph shows Force $F(N)$ on the vertical axis and distance $d(m)$ on the horizontal axis. The force is zero for d from 0 to 3. It then jumps to 2 N and remains constant until $d = 12$. From $d = 12$ to $d = 15$, the force decreases linearly to 0 N. Dashed lines indicate the constant force region and the final zero force point.



6. Which of the following pairs can never give equal numerical values of temperature?
- a. Kelvin and Fahrenheit b. Fahrenheit and Celsius
c. Kelvin and Celsius d. Reaumur and Rankine
7. How far should an object be held from a concave mirror of focal length 40 cm so as to obtain a real image twice the size of the object?
- a. 10 cm b. 20 cm c. 30 cm d. 60 cm
8. The refractive index of glass slab is $\frac{5}{3}$ and that of water is $\frac{4}{3}$. What is the critical angle for a ray of light passing from glass to water?
- a. $\sin^{-1}(\frac{3}{4})$ b. $\sin^{-1}(\frac{3}{5})$ c. $\sin^{-1}(\frac{4}{5})$ d. none
9. If a unit charge is taken from one point to another over an equipotential surface, then
- a. work is done on the charge b. work is done by the charge
c. work on the charge is constant d. no work is done
10. An electron of charge 'e' is at rest in an electric field between two plates separated by distance 'd' and with potential difference 'V', then find the force experienced by it
- a. Fehler! b. Fehler! c. Fehler! d. Fehler!
11. In a charged capacitor the energy resides
- a. on the field between the plates
b. on the positive plate
c. around the edge of capacitor plates
d. on both positive and negative plates

Group – B

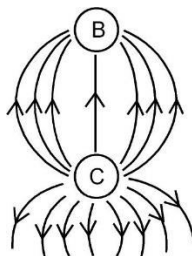
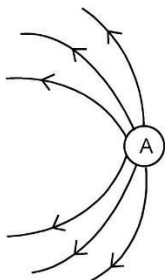
Give short answer to the following questions.

[8 × 5 = 40]

12. a) Resultant of two equal forces may have the magnitude equal to one of the forces. At what angle between the two equal forces, is this possible? [1]
- b) Distance travelled by a particle at any instant 't' can be represented as $s = A(t+B) + Ct^2$. Find the dimensions of A and C. [2]

- c) Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{C} = \vec{A} + \vec{B}$ and $C = A - B$. Find angle between $\vec{A}$ and $\vec{B}$. [2]
13. a) Using v-t graph, derive the relation $v^2 = u^2 + 2as$, where the symbols carry their usual meaning. [2]
- b) A car travelling with a speed of 15 m/s is braked and it slows down with uniform retardation. It covers a distance of 88 m as its velocity reduces to 7 m/s. Calculate:
- i) The retardation of the car. [1]
- ii) Further distance travelled by the car before coming to rest if it continues to slow down at the same rate. [1]
- iii) Sketch the nature of velocity-time graph for motion of car. [1]
14. a) Show that total mechanical energy of a falling body is conserved when it moves under a gravitational field. [2]
- b) A 650 KW power engine of a vehicle of 1.5×10^5 kg is rising on an inclined plane of inclination 1 in 100 with a constant speed of 60Km/hr. Find the frictional force between the wheels of the vehicles and the plane. [3]
15. a) Explain why the possibility of "water taps burst" in severe winter. [1]
- b) Define the coefficient of cubical expansion of a solid and hence write an expression for the variation of its density with temperature. [1+1]
- c) A glass flask with volume 200 cm^3 is filled to the brim with mercury at 20°C . How much mercury overflows when the temperature of the system is raised to 100°C .
 $(\alpha \text{ for glass} = 1.8 \times 10^{-6} / ^\circ\text{C}, \gamma \text{ for mercury} = 1.8 \times 10^{-4} / ^\circ\text{C})$ [2]
16. a) What is specific latent heat of fusion of ice? [1]
- b) Develop an expression for the determination of the latent heat of fusion of ice. [2]
- c) A copper calorimeter of mass 300 gm contains 500 gm of water at temperature 15°C . A 560 gm block of aluminium at temperature 100°C is dropped in the calorimeter and the temperature is observed to increase to 25°C . Find the specific heat capacity of aluminium. (Sp. Heat capacity of water = $1 \text{ cal gm}^{-1} ^\circ\text{C}^{-1}$, Sp. Heat capacity of copper = $0.093 \text{ cal gm}^{-1} ^\circ\text{C}^{-1}$) [2]

17. a) Define 'concave mirror' and state one daily application of it. [2]
 b) What do you mean by critical angle and total internal reflection? [1]
 c) What is the apparent position of an object below a rectangular block of glass 6 cm thick if a layer of water 4 cm thick is on the top of the glass? (Refractive index of glass = **Fehler!** and of water = **Fehler!**) [2]
18. a) Derive the relation between refractive index, minimum deviation and angle of prism. [3]
 b) A glass prism is immersed in water. What will happen to the angle of minimum deviation? [2]
19. a) What is meant by quantization of charge? [1]
 b) Why are sharp edges or points strictly avoided in electrical machines? [2]
 c) Figure shows the electric lines around three charges A, B and C.



- i. Which charges are positive? [1]
 ii. Which charge has the largest magnitude? Why? [1]

OR

- a) Define capacitance of a capacitor. A parallel plate capacitor has vacuum in the space between the plates. If we double the amount of charge on each plate, what happens to the capacitance? [2]
 b) A hollow spherical conductor of radius 12 cm is charged to $6 \times 10^{-6} \text{C}$. Find the electric field strength at the surface of sphere, inside the sphere at 8 cm and at distance 15 cm from the sphere. ($\epsilon_0 = 8.86 \times 10^{-12} \text{C}^2 \text{N}^{-1} \text{m}^{-2}$) [3]

Group – C

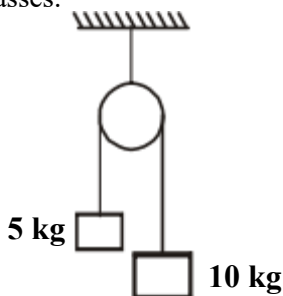
Give long answer to the following questions.

[3 × 8 = 24]

20. a) Show that the trajectory of a projectile projected from ground making an angle ' θ ' with the horizontal is parabolic in nature. [2]
- b) A base ball is thrown towards a player with an initial velocity 20 m/s and 45° with the horizontal. At the moment the ball is thrown, the player is 50m from the thrower, at what speed and in what direction must he run to catch the ball at the same height at which it was released? [3]
- c) Can a body have a constant speed but changing velocity? Explain with example. [1]
- d) An object is dropped from the top of the tower of height 156.8 m and at the same time another object is thrown vertically upward with a velocity of 78.1 m/s from the foot of the tower. When and where will the two objects meet? [2]

OR

- a) What is an angle of repose? [1]
- b) Show that the coefficient of static friction is equal to the tangent of angle of repose. [2]
- c) Why is it difficult to run fast on sand? [1]
- d) An iron block of mass 10 kg rests on a wooden plane inclined at 30° to the horizontal. It is found that least force parallel to the plane which causes the block to slide up the plane is 100 N. Calculate the coefficient of sliding friction between wood and iron. [2]
- e) Two masses as shown in the figure are suspended from a massless pulley. Calculate: ($g = 10 \text{ ms}^{-2}$)
- (i) tension in the string.
- (ii) acceleration of the masses. [2]



21. a) Define temperature gradient. [1]
 b) Relate it with rate of heat transfer along a conductor. [2]
 c) Calculate the quantity of heat conducted through 2m^2 on a brick wall 3 cm thick in 1 hour if the temperature on one side is 5°C and on the other side is 25°C . [Thermal conductivity of brick = $0.12\text{ W m}^{-1}\text{ K}^{-1}$] [2]
 d) The length of an iron rod is measured by a brass scale. When both of them are at 10°C , the measured length is 50cm. What is the length of the rod at 40°C when measured by the brass scale at 40°C ? (α for brass = $24 \times 10^{-6}^\circ\text{C}^{-1}$, α for iron = $16 \times 10^{-6}^\circ\text{C}^{-1}$) [3]
22. a) Define electric potential at a point due to point charge and derive its expression. [1+2]
 b) How can a number of capacitors be connected to increase and decrease the effective capacitance? Find the respective expressions for the effective capacitance in each case. [3]
 c) Two capacitors, of capacitances $4\text{ }\mu\text{F}$ and $12\mu\text{F}$ respectively, are connected in series and the combination connected momentarily across a 200 V battery. The charged capacitors are now isolated and connected in parallel, similar charged plates connected together. What would be the resulting potential difference across the combinations? [2]

❧ - The End - ❧